Open Source Audit Project

Student Name: Manmeet Walmik Sonawane

Registration Number: 24BAI10180

Course: Open Source Software

Chosen Software: Git

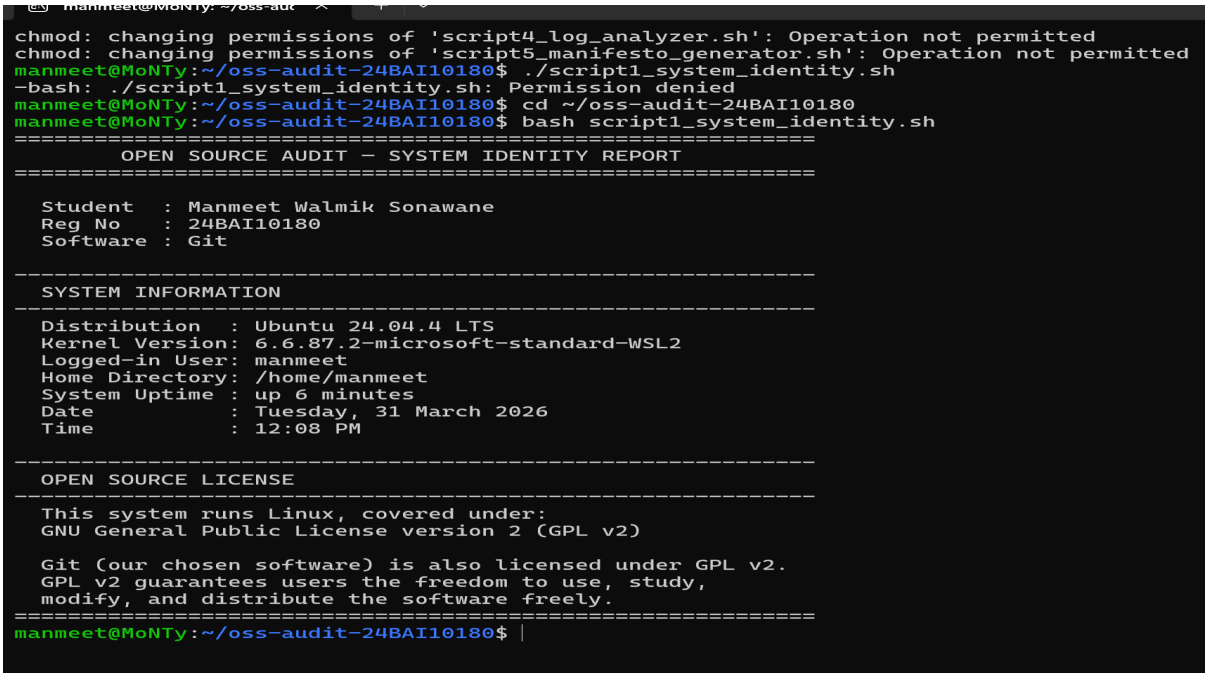
Part A — Introduction and Philosophy

This project explores Git, an open-source version control system created by Linus Torvalds in 2005. It was developed as a response to the limitations of proprietary tools and has since become the backbone of modern software development.

Part B — Linux Implementation

Git was installed and tested on Ubuntu Linux. Scripts were executed to demonstrate system interaction, automation, and open-source concepts.

Part C — Script Outputs



```
chmod: changing permissions of 'script4_log_analyzer.sh': Operation not permitted
chmod: changing permissions of 'script5_manifesto_generator.sh': Operation not permitted
manmeet@MoNTy:~/oss-audit-24BAI10180$ ./script1_system_identity.sh
-bash: ./script1_system_identity.sh: Permission denied
manmeet@MoNTy:~/oss-audit-24BAI10180$ cd ~/oss-audit-24BAI10180
manmeet@MoNTy:~/oss-audit-24BAI10180$ bash script1_system_identity.sh
============================================================
         OPEN SOURCE AUDIT — SYSTEM IDENTITY REPORT
============================================================

  Student  : Manmeet Walmik Sonawane
  Reg No   : 24BAI10180
  Software : Git

------------------------------------------------------------
  SYSTEM INFORMATION
------------------------------------------------------------
  Distribution  : Ubuntu 24.04.4 LTS
  Kernel Version: 6.6.87.2-microsoft-standard-WSL2
  Logged-in User: manmeet
  Home Directory: /home/manmeet
  System Uptime : up 6 minutes
  Date          : Tuesday, 31 March 2026
  Time          : 12:08 PM

------------------------------------------------------------
  OPEN SOURCE LICENSE
------------------------------------------------------------
  This system runs Linux, covered under:
  GNU General Public License version 2 (GPL v2)

  Git (our chosen software) is also licensed under GPL v2.
  GPL v2 guarantees users the freedom to use, study,
  modify, and distribute the software freely.
============================================================
manmeet@MoNTy:~/oss-audit-24BAI10180$
```

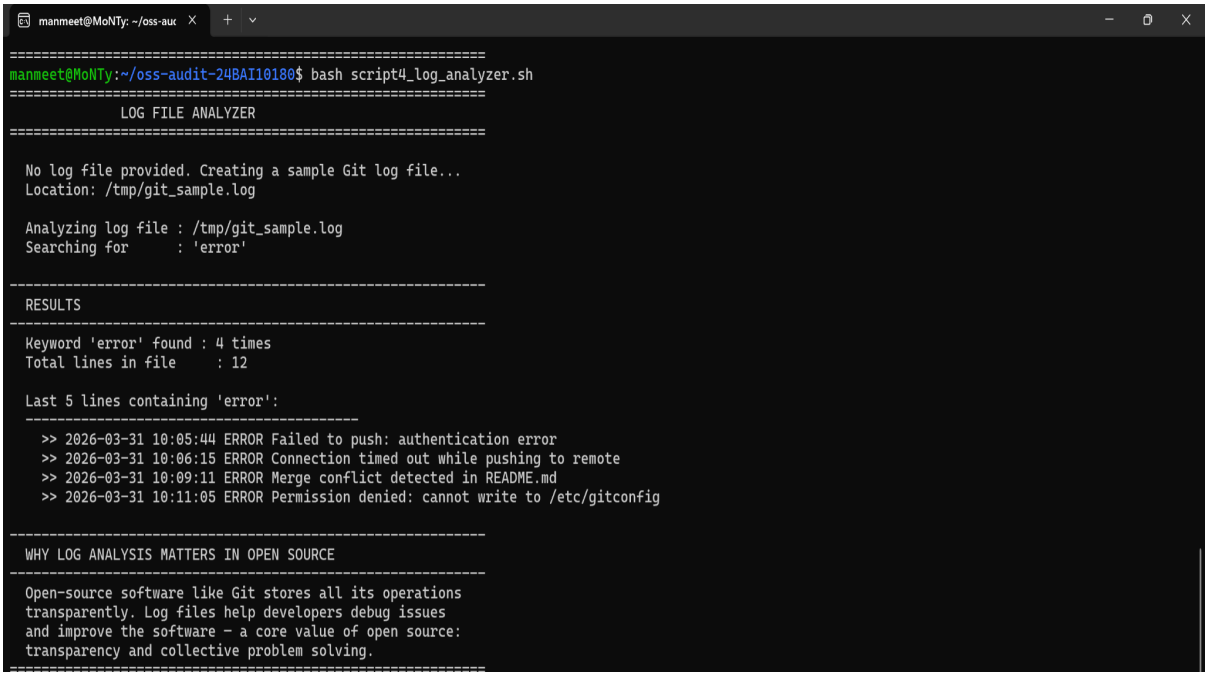
Figure 1: System Identity Script Output

```
manmeet@MoNTy: ~/oss-auc

==============================================================
manmeet@MoNTy:~/oss-audit-24BAI10180$ bash script2_package_inspector.sh
==============================================================
          FOSS PACKAGE INSPECTOR
==============================================================

  [OK] 'git' is installed on this system.

  Version   : git version 2.43.0

  Package details (dpkg):
  Installed : git 1:2.43.0-1ubuntu7.3

  Git License   : GPL v2 (GNU General Public License version 2)
  Official Site : https://git-scm.com
  Source Code   : https://github.com/git/git

--------------------------------------------------------------
  OPEN SOURCE PHILOSOPHY NOTES
--------------------------------------------------------------

  git      - Built by Linus Torvalds in 2005 when BitKeeper
             revoked free access. Git proved that open-source
             communities can build better tools than corporations.
  httpd    - Apache HTTP Server: the web server that powered
             the early internet, built entirely by volunteers.
  mysql    - MySQL: open source at the heart of millions of apps,
             a dual-license model showing OSS and commerce can coexist.
  vlc      - VLC: born in a French university, plays anything,
             shared freely so everyone gets access to media.
  firefox  - Firefox: a nonprofit fighting for an open, private web
             against corporate browser monopolies.
  python3  - Python: shaped entirely by community contributions,
             the most beginner-friendly open-source language.
```

Figure 2: Package Inspector Output

```
manmeet@MoNTy: ~/oss-auc
==========================================================
manmeet@MoNTy:~/oss-audit-24BAI10180$ bash script3_disk_auditor.sh
==========================================================
         DISK AND PERMISSION AUDITOR
==========================================================

  Directory            Permissions (type user group) Size
  ----------------------------------------------------------
  /etc                 drwxr-xr-x root root      4.3M
  /var/log             drwxrwxr-x root syslog    33M
  /home                drwxr-xr-x root root      80K
  /usr/bin             drwxr-xr-x root root      95M
  /tmp                 drwxrwxrwt root root      20K

----------------------------------------------------------
  GIT CONFIG DIRECTORY CHECK (Software-specific audit)
----------------------------------------------------------

  Git global config (/home/manmeet/.gitconfig) not found.
  This means Git has not been configured yet on this system.
  Run: git config --global user.name 'Your Name'

  No system-wide Git config found at /etc/gitconfig

----------------------------------------------------------
  WHY PERMISSIONS MATTER IN OPEN SOURCE
----------------------------------------------------------
  File permissions in Linux control who can read, write,
  or execute a file. Open-source software respects these
  boundaries - Git stores user configs in home directories
  (private) and system configs in /etc (admin-controlled).
----------------------------------------------------------
```

Figure 3: Disk Auditor Output



```
manmeet@MoNTy: ~/oss-auc
==============================================================
manmeet@MoNTy:~/oss-audit-24BAI10180$ bash script4_log_analyzer.sh
==============================================================
              LOG FILE ANALYZER
==============================================================

  No log file provided. Creating a sample Git log file...
  Location: /tmp/git_sample.log

  Analyzing log file : /tmp/git_sample.log
  Searching for      : 'error'

--------------------------------------------------------------
  RESULTS
--------------------------------------------------------------
  Keyword 'error' found : 4 times
  Total lines in file     : 12

  Last 5 lines containing 'error':
  ----------------------------------------
    >> 2026-03-31 10:05:44 ERROR Failed to push: authentication error
    >> 2026-03-31 10:06:15 ERROR Connection timed out while pushing to remote
    >> 2026-03-31 10:09:11 ERROR Merge conflict detected in README.md
    >> 2026-03-31 10:11:05 ERROR Permission denied: cannot write to /etc/gitconfig

--------------------------------------------------------------
  WHY LOG ANALYSIS MATTERS IN OPEN SOURCE
--------------------------------------------------------------
  Open-source software like Git stores all its operations
  transparently. Log files help developers debug issues
  and improve the software — a core value of open source:
  transparency and collective problem solving.
==============================================================
```

Figure 4: Log Analyzer (Sample Log)

```
manmeet@MoNTy: ~/oss-auc

  Open-source software like Git stores all its operations
  transparently. Log files help developers debug issues
  and improve the software — a core value of open source:
  transparency and collective problem solving.
==========================================================
manmeet@MoNTy:~/oss-audit-24BAI10180$ bash script4_log_analyzer.sh /var/log/syslog error
==========================================================
              LOG FILE ANALYZER
==========================================================

  Analyzing log file : /var/log/syslog
  Searching for      : 'error'

----------------------------------------------------------
  RESULTS
----------------------------------------------------------
  Keyword 'error' found : 12 times
  Total lines in file     : 1479

  Last 5 lines containing 'error':
  ----------------------------------------
    >> 2026-03-31T10:10:22.026558+00:00 MoNTy systemd[1]: apport-autoreport.timer - Process error reports when automatic reporting is enabled (timer
 based) was skipped because of an unmet condition check (ConditionPathExists=/var/lib/apport/autoreport).
    >> 2026-03-31T10:10:22.027801+00:00 MoNTy kernel: RAS: Correctable Errors collector initialized.
    >> 2026-03-31T10:11:08.014467+00:00 MoNTy kernel: WSL (325) ERROR: CheckConnection: getaddrinfo() failed: -5
    >> 2026-03-31T12:10:19.265190+00:00 MoNTy kernel: WSL (325) ERROR: CheckConnection: getaddrinfo() failed: -5
    >> 2026-03-31T12:11:54.421680+00:00 MoNTy kernel: WSL (325) ERROR: CheckConnection: getaddrinfo() failed: -5

----------------------------------------------------------
  WHY LOG ANALYSIS MATTERS IN OPEN SOURCE
----------------------------------------------------------
  Open-source software like Git stores all its operations
  transparently. Log files help developers debug issues
  and improve the software — a core value of open source:
  transparency and collective problem solving.
```

Figure 5: Log Analyzer (System Log)

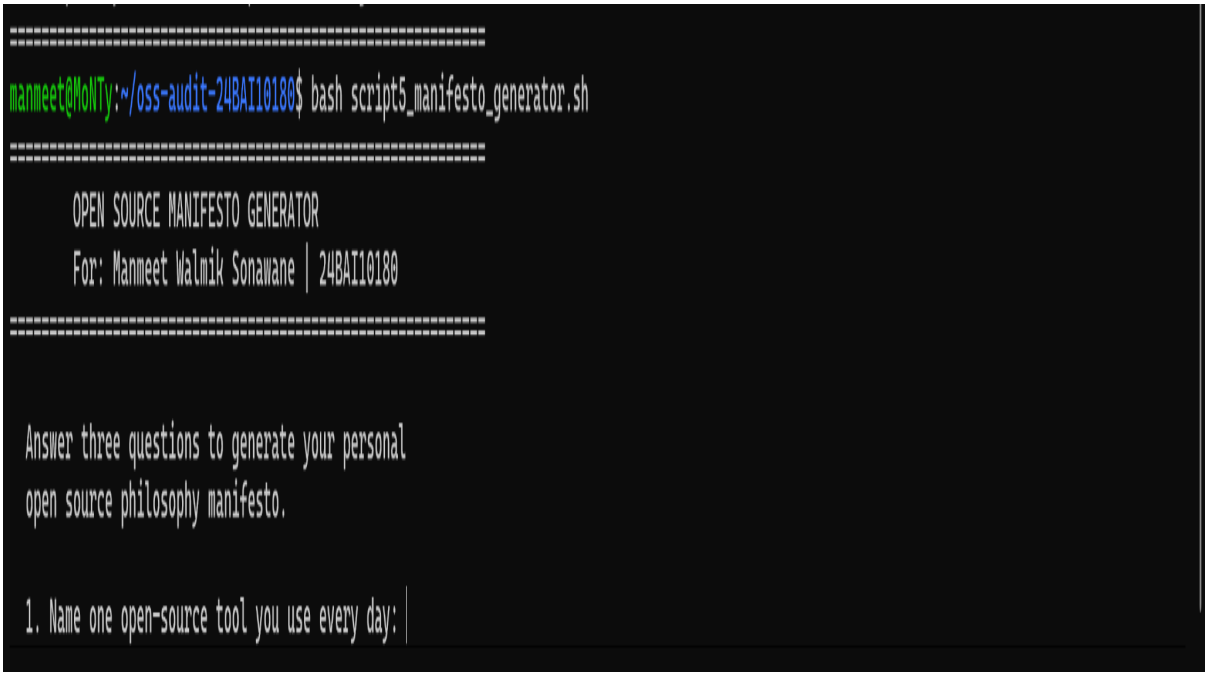


Figure 6: Manifesto Generator Input

```
manmeet@MoNTy: /tmp
manmeet@MoNTy:~/oss-audit-24BAI10180$ cd /tmp
manmeet@MoNTy:/tmp$ cp ~/oss-audit-24BAI10180/script5_manifesto_generator.sh .
manmeet@MoNTy:/tmp$ bash script5_manifesto_generator.sh
==============================================================
         OPEN SOURCE MANIFESTO GENERATOR
         For: Manmeet Walmik Sonawane | 24BAI10180
==============================================================

  Answer three questions to generate your personal
  open source philosophy manifesto.

  1. Name one open-source tool you use every day: git
  2. In one word, what does 'freedom' mean to you? freedom
  3. Name one thing you would build and share freely: app

  Generating your manifesto...

  Manifesto saved to: manifesto_manmeet.txt

==============================================================
==============================================================
          MY OPEN SOURCE MANIFESTO
==============================================================

  Author   : Manmeet Walmik Sonawane (24BAI10180)
  Date     : 31 March 2026 at 12:49 PM
  System   : manmeet@MoNTy

--------------------------------------------------------------

  Every day, I rely on git — a tool that exists
  not because a corporation decided to sell it, but because
  someone chose to build it openly and share it freely.

  To me, freedom means freedom. In the world of software,
  that freedom is not given — it is built, line by line,
  by communities who believe that knowledge belongs to everyone.

  If I could contribute one thing to that world, I would
  build app and share it freely — because the software
  I depend on today exists because someone before me made
  the same choice.

  Open source is not just a license. It is a philosophy:
  that we rise together when we share what we know.
```

Figure 7: Manifesto Generated Output (Part 1)

```
============================================================
         MY OPEN SOURCE MANIFESTO
============================================================

  Author  : Manmeet Walmik Sonawane (24BAI10180)
  Date    : 31 March 2026 at 12:49 PM
  System  : manmeet@MoNTy

------------------------------------------------------------

  Every day, I rely on git — a tool that exists
  not because a corporation decided to sell it, but because
  someone chose to build it openly and share it freely.

  To me, freedom means freedom. In the world of software,
  that freedom is not given — it is built, line by line,
  by communities who believe that knowledge belongs to everyone.

  If I could contribute one thing to that world, I would
  build app and share it freely — because the software
  I depend on today exists because someone before me made
  the same choice.

  Open source is not just a license. It is a philosophy:
  that we rise together when we share what we know.

------------------------------------------------------------
  Chosen Software for OSS Audit : Git
  Git License                   : GPL v2
  This manifesto was generated on 31 March 2026
============================================================
manmeet@MoNTy:/tmp$
```

Figure 8: Manifesto Generated Output (Part 2)

Conclusion

This project demonstrates the power of open-source software through Git and Linux-based scripting. The scripts highlight automation, transparency, and efficiency — key principles of open source.